

PAPER CASE 03 / BIOTECH ATLAS

SINGLE-CELL

CAR T ATLAS

How a persistent clinical phenotype
becomes a cell-state hypothesis.

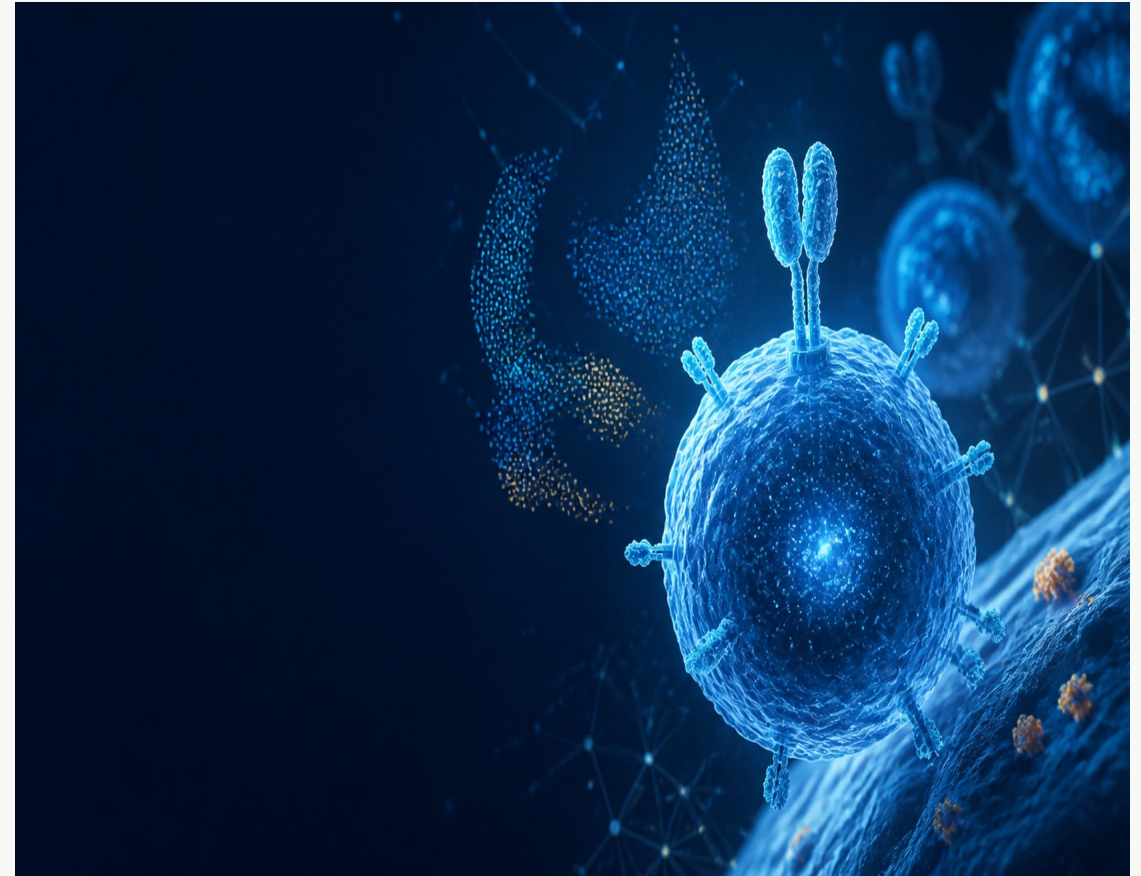
Bai et al. · Nature 634 · 702–711 · 2024

82 patients · 695,819 analyzed cells

CONCEPTUAL COVER VISUAL

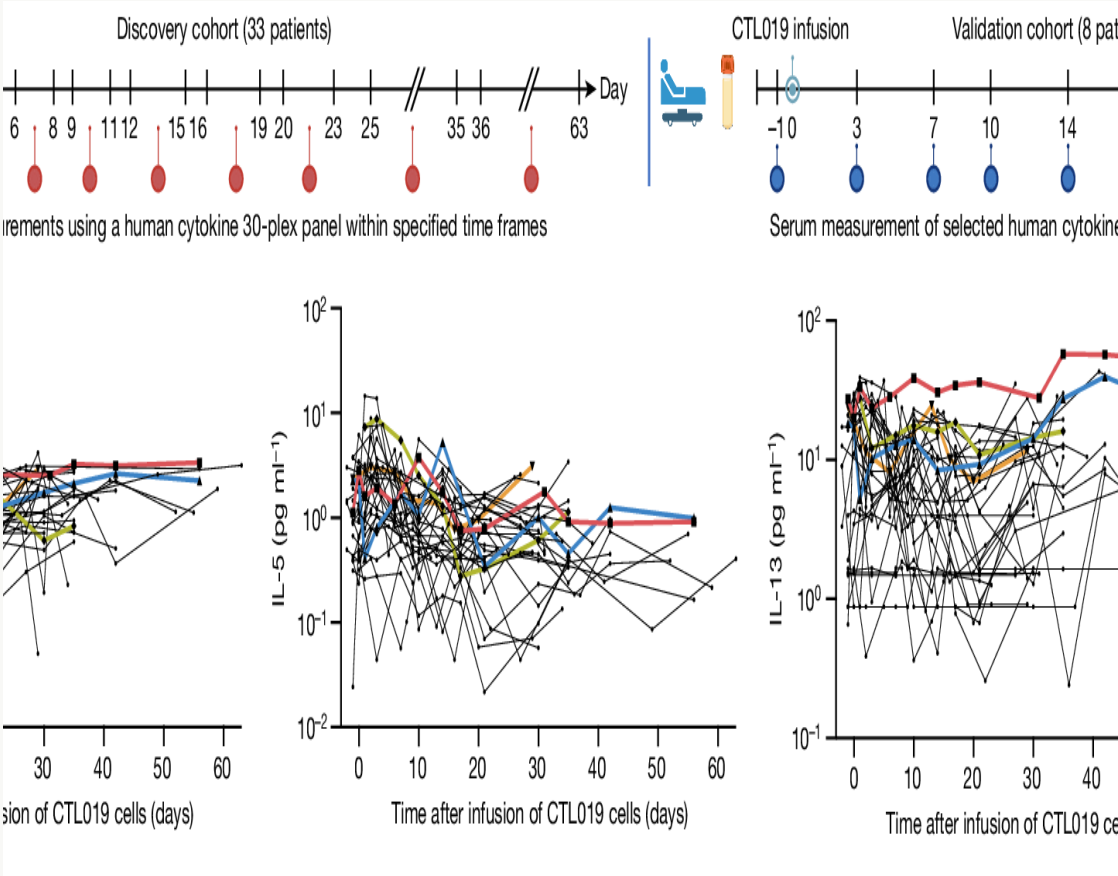
CLINICAL / CELLULAR / PRECLINICAL

DOI 10.1038/s41586-024-07762-w



01 / CLINICAL PERSISTENCE

The clinical signal is unusually long-lived
Before asking what the cell is, establish how long it remains clinically visible.
2 / 12
ATLAS / 03
Source figure / Fig. 4a–b · longitudinal serum measurements
PERSISTENCE READOUT
8.4
years
median BCA-L duration
The phenotype is not a single
measurement—it is a longitudinal trace.
OBSERVATION
long B-cell aplasia
QUESTION
what sustains the state?
NEXT
single-cell atlas
Evidence anchor: BCA-L group median BCA duration 8.4 years. Clinical
persistence is the entry point, not the conclusion.

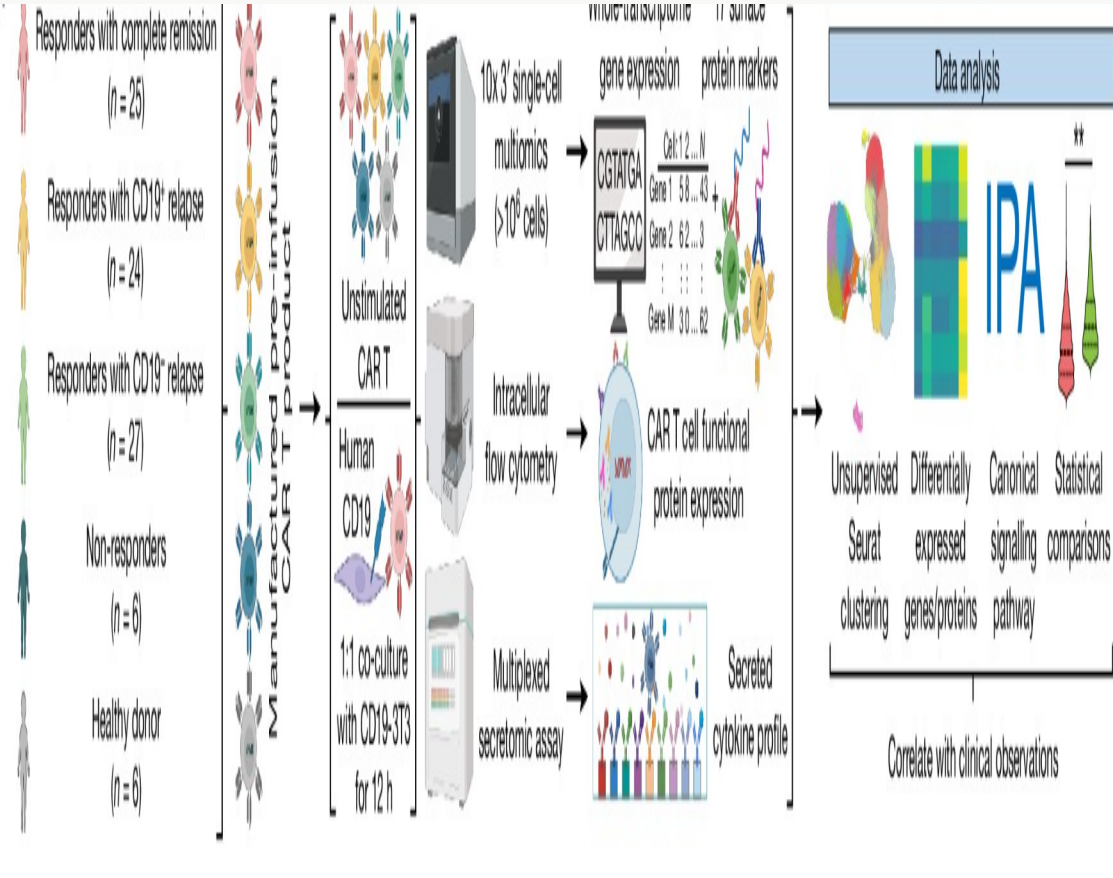


02 / STUDY DESIGN

One cohort becomes a multi-omic research system
The paper connects patient persistence to a measured cell-state atlas through five linked readout layers.

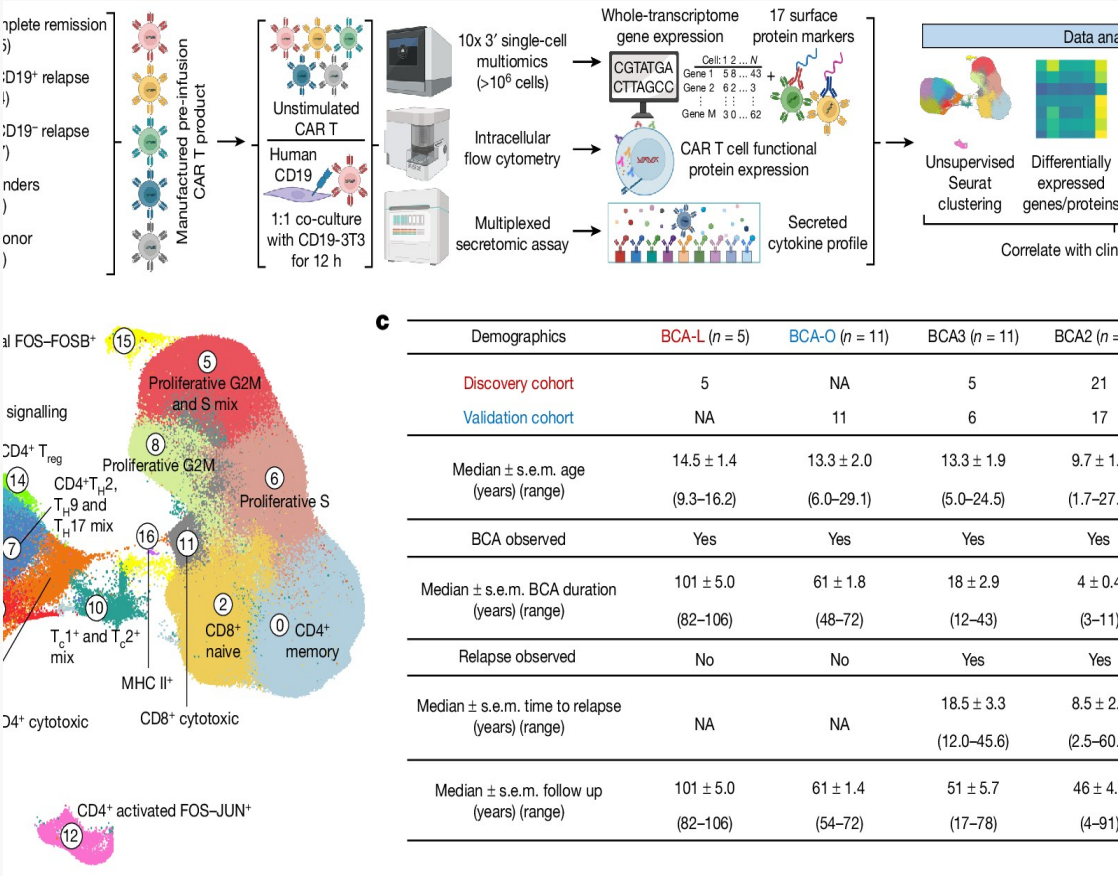
3 / 12
ATLAS / 03
01
PATIENT
COHORT
82 ALL patients
+ 6 healthy donors
02
CAR T
PRODUCT
manufactured
CAR T cells
03
SINGLE-CELL
MULTIOMICS
10x 3' assay
>10⁶ cells
04
FUNCTIONAL
READOUTS
flow · secretomics
ATAC
05
CLINICAL
CORRELATION
17 clusters
Independent neutral baseline · same case content and page count
BCA strata

Source figure / Fig. 1a · cohort → measurement layers → data analysis



03 / CELL ATLAS

695,819 cells resolve into 17 states
The UMAP is the map from which the later mechanistic claims depart.
4 / 12
ATLAS / 03
Fig. 1b / UMAP + cluster annotation
695,819
analyzed cells
17
cell states
ATLAS READOUT / FOUR STATE FAMILIES
NAIVE
CYTOTOXIC
REGULATORY
PROLIFERATIVE
Design principle: preserve the paper’s cluster colors; use blue annotations to explain the map.
Source figure / Fig. 1b. Cell labels and original color encoding are preserved.



04 / PERSISTENCE

Persistence is a gradient, not a binary

BCA strata create the clinical axis reused across the paper.

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ATLAS / 03

Fig. 1c / demographics and follow-up

PERSISTENCE GRADIENT

BCA-L

101

BCA-O

61

BCA3

18

BCA2

4

BCA1

1

BCA-L / LONGEST

n = 5 · 101 months

no relapse observed

BCA1 / SHORTEST

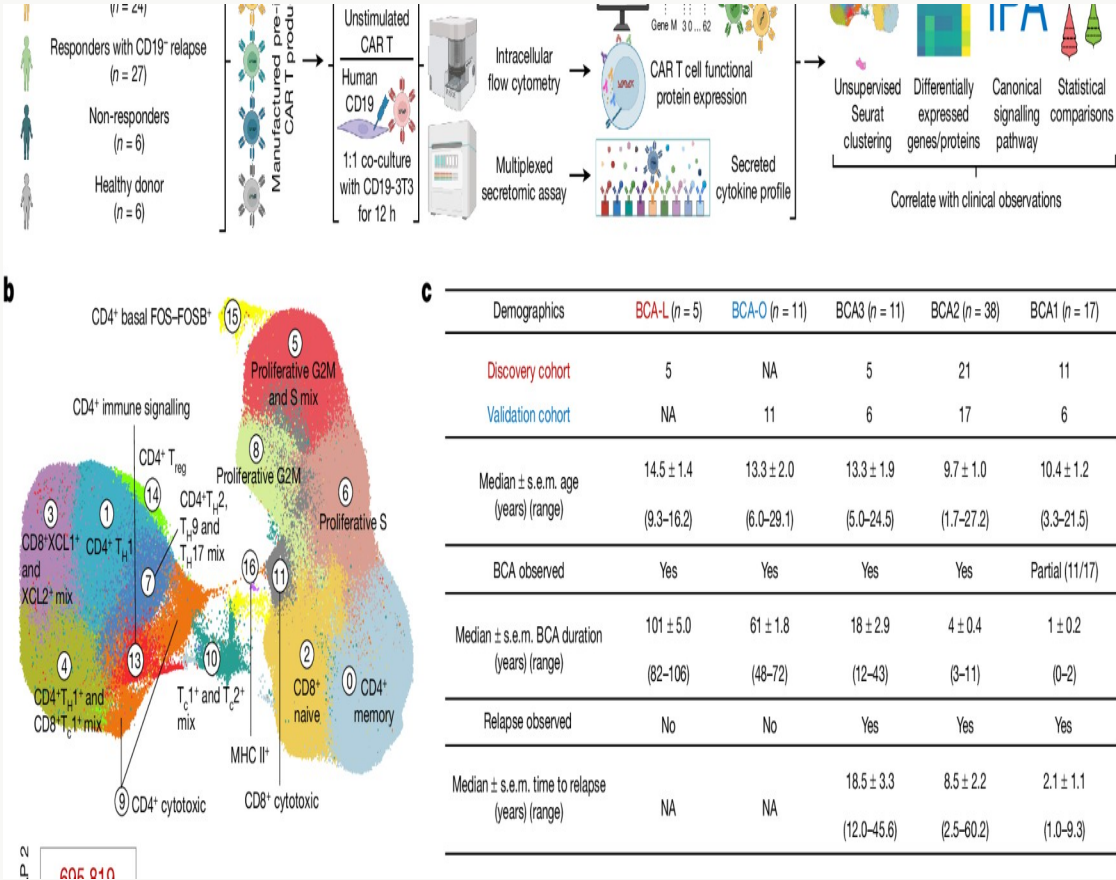
n = 17 · 1 month

relapse observed

INTERPRETATION

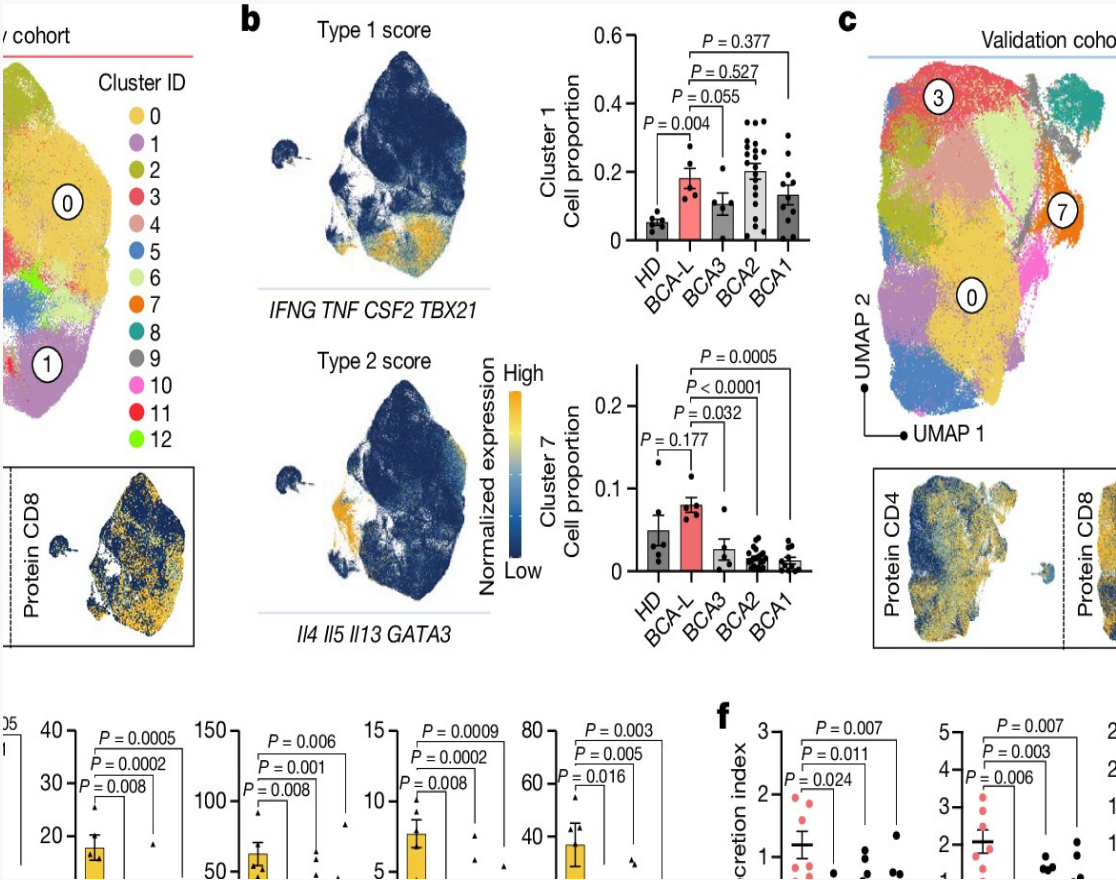
Persistence is a continuum with a clinical gradient, not a yes/no label.

Source figure / Fig. 1c. Duration values shown are the paper’s reported values; units are months in the source table.



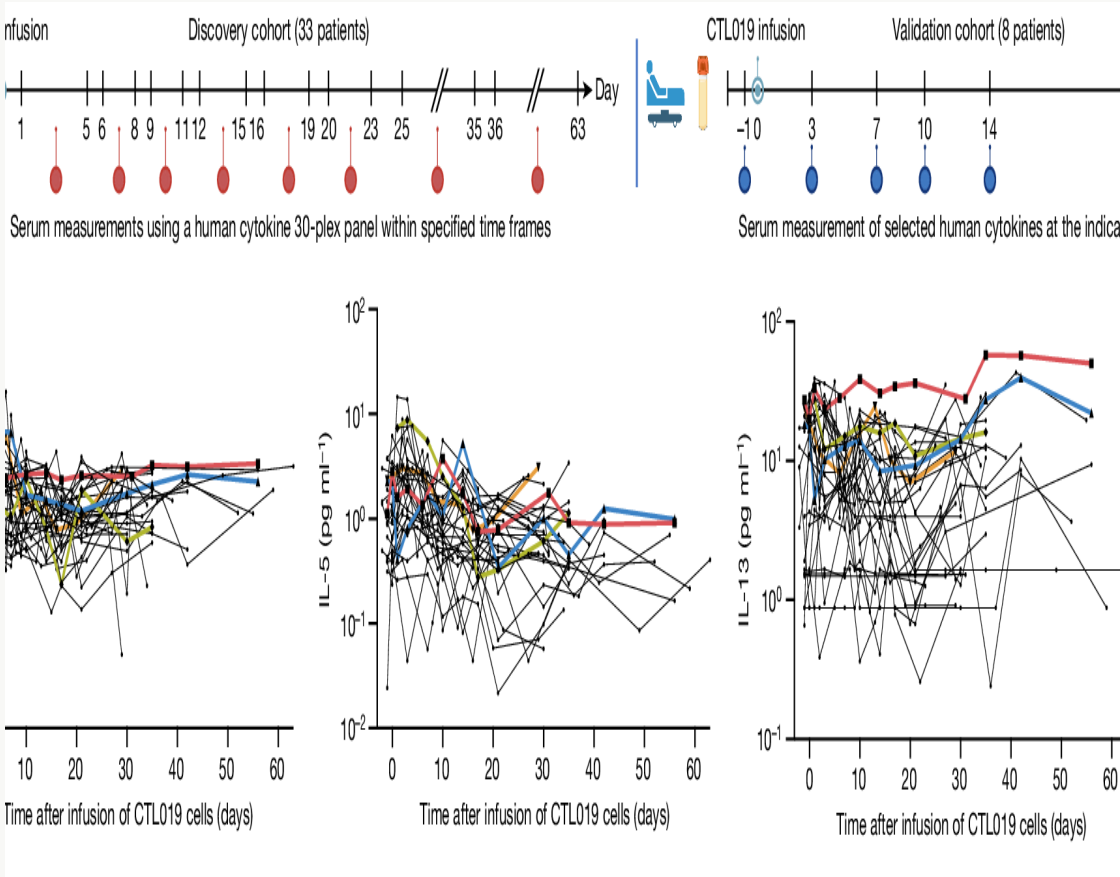
05 / TYPE 2 STATE

The long-remission state reads as type 2
Four orthogonal readouts point to the same biological state.
6 / 12
ATLAS / 03
Fig. 2a–d / type 2 score
Fig. 2e–f / secretion
Fig. 2g–j / ATAC + GATA3
PERTURBATION
GATA3 / STAT6
repeat stimulation
→ tumour-cell lysis
TYPE 2 SIGNATURE
IL4 · IL5 · IL13
GATA3
enriched in the long-remission group
EVIDENCE RULE / CONVERGENCE
TRANSCRIPTOME
SECRETION
CHROMATIN
→ one coherent type 2 state
Source figure / Fig. 2. Direct paper panels used as evidence modules;
interpretation is newly organized.



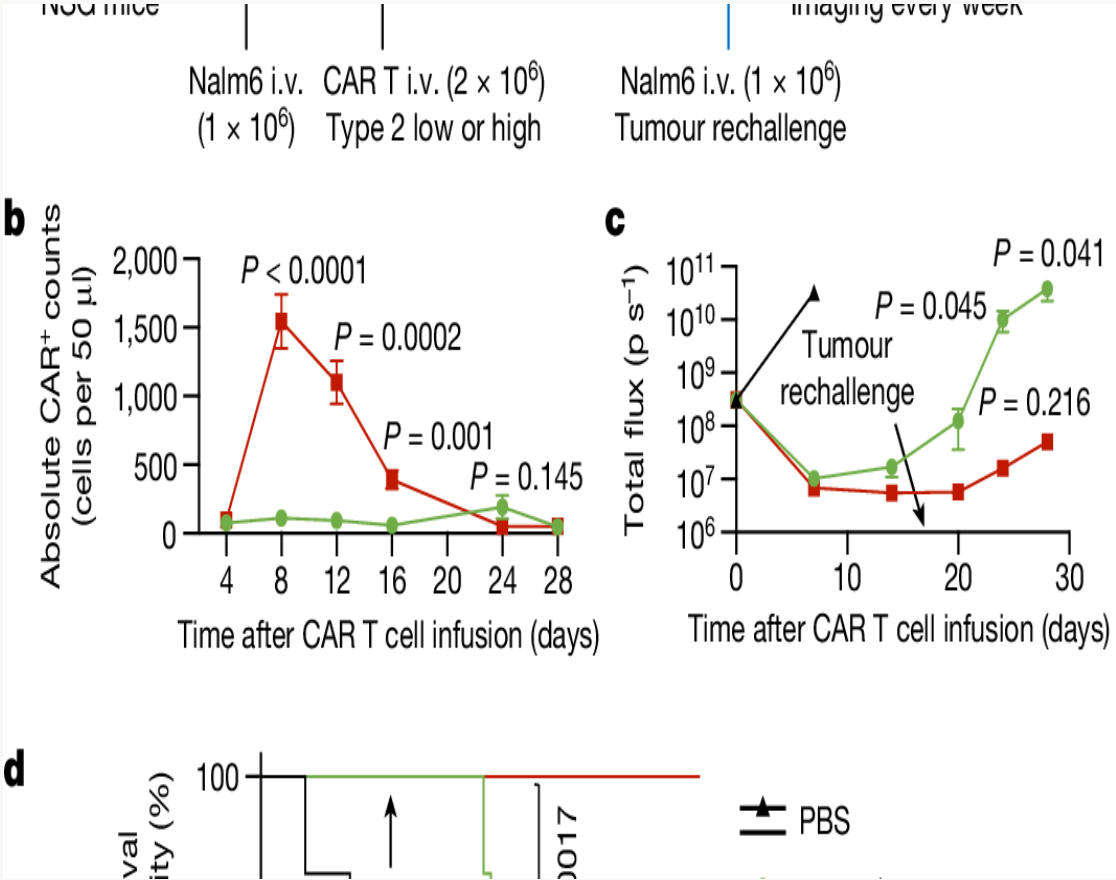
07 / SECRETOME

The serum trace keeps the biology longitudinal
A state is more credible when it survives time, not just a snapshot.
8 / 12
ATLAS / 03
Fig. 4a–b / collection timeline + time series
345
measurements
30
cytokines
33
patients
Fig. 4c–e / cytokine windows + heatmap
TYPE 2 WINDOW
IL-4 · IL-5 · IL-13
serial evidence across days 1–63
33 discovery + 8 validation patients
Source figure / Fig. 4. Longitudinal panels are retained; the surrounding blue
modules explain scale and timing.



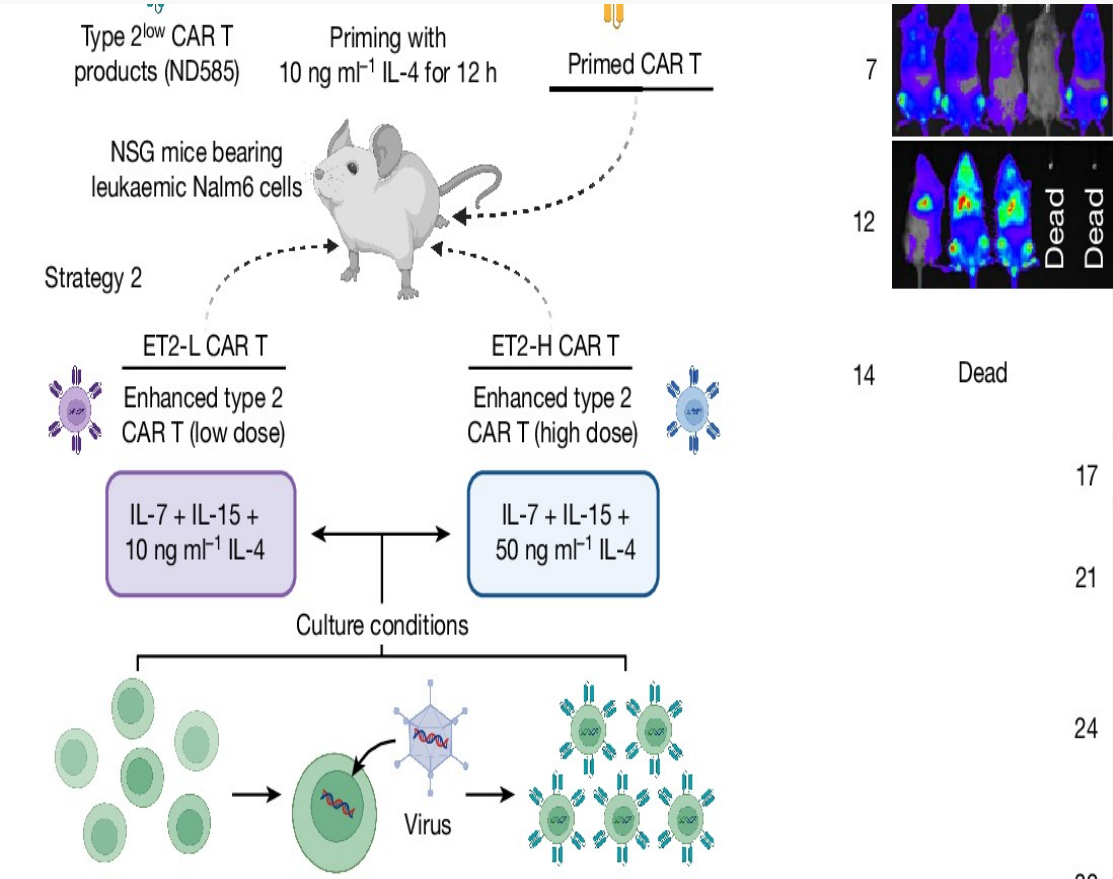
08 / IN VIVO TEST

Type 2-high cells pass the recall test
The state is tested as function, not merely described as phenotype.
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ATLAS / 03
Fig. 5a–c / mouse sequence + expansion
Fig. 5d / survival after rechallenge
IN VIVO SEQUENCE
–7
Nalm6
0
CAR T
17
rechallenge
24–28
imaging
READOUT
Type 2-high cells expand more and show stronger recall in the reported model;
selected readouts n=5 mice.
EVIDENCE BOUNDARY
Preclinical model supports function; it does not establish clinical efficacy.
Source figure / Fig. 5. Experimental evidence is shown with the preclinical
boundary kept explicit.



09 / ENGINEERING

IL-4 turns the state into a design lever
The final figure converts a cell-state hypothesis into a manufacturing strategy.
10 / 12
ATLAS / 03
Fig. 6a / priming and ET2-L/H strategies
Fig. 6b / in vivo imaging
10
ng ml⁻¹
50
ng ml⁻¹
Fig. 6c–d / expansion + survival
DESIGN OPTIONS
priming: 10 ng ml⁻¹ / 12 h
ET2-L: 10 · ET2-H: 50
controlled manufacturing exposure
INTERPRETATION
Promising engineering lever
still bounded by model and study design
Source figure / Fig. 6. Concentrations and group labels remain tied to the paper; the design framing is editorialized.



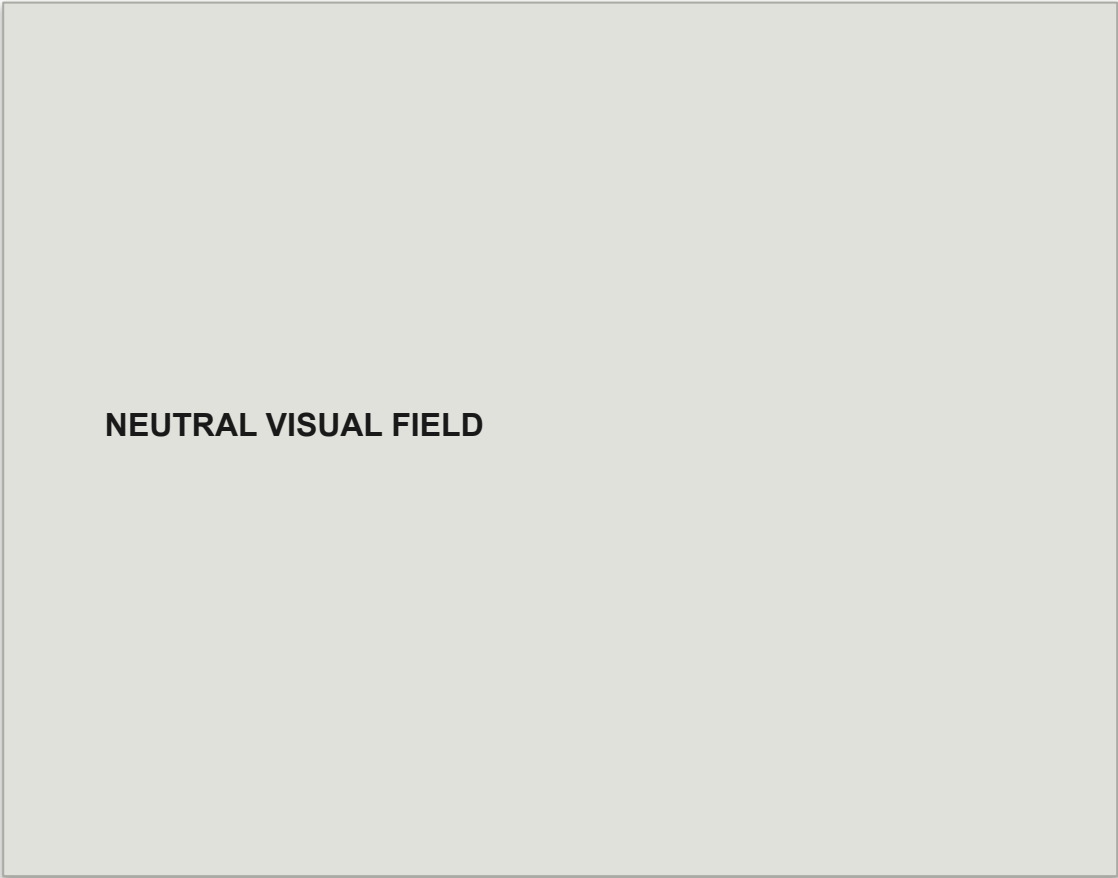
10 / EVIDENCE CHAIN

The argument is strongest when each layer stays distinct
Do not collapse clinical association, mechanism and intervention into one claim.

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ATLAS / 03
01
OBSERVED
CLINICAL
8.4-year persistence
Fig. 1 / 4
02
RESOLVED
CELLULAR
type 2 state
Fig. 2
03
PROPOSED
MECHANISM
cluster 2 network
Fig. 3
04
TESTED
PRECLINICAL
recall + IL-4
Fig. 5 / 6
READING RULE
Clinical persistence anchors the story; cellular evidence resolves the state;
mechanism remains a candidate; preclinical work tests the direction.

Evidence boundary: association → state definition → candidate explanation →
Independent neutral baseline · same case content and page count
functional test / engineering lever

A rigorous research presentation makes evidence type and evidence boundary



THE TAKEAWAY

Type 2 function
may sustain fitness.
A durable clinical phenotype
becomes a recognizable cell state,
a candidate circuit and an engineering lever.
A strong research story shows its figures—and its limits.

Fig. 2
Fig. 3
Fig. 6 / engineering strategy
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